

TELECOM SUBSCRIBER IDENTITY PLATFORM

Resolved a multi-account identification failure across API, cache, and data-pipeline layers

ROLE

Solutions Architect / Cloud & Data Engineer

IMPACT

Customer identification, delivery recovery, performance, production readiness

THE CHALLENGE

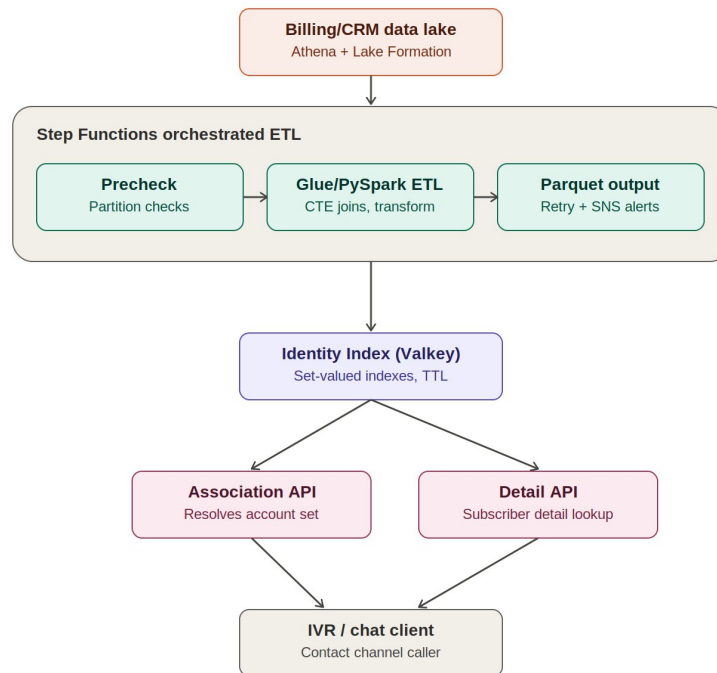
CUSTOMER IMPACT

The service returned only the first account associated with a caller, producing incomplete identification when phone or personal ID mapped to multiple accounts.

DELIVERY CONSTRAINT

Poorly documented fixed/mobile billing data blocked the external team's convergent-table extraction and joins.

ARCHITECTURE AT A GLANCE



DECISIONS THAT CHANGED THE OUTCOME

1 MODEL THE REAL CARDINALITY

Made account ID canonical and phone/personal ID set-valued secondary indexes so shared identifiers return every associated account.

2 DISAMBIGUATE BEFORE DETAIL

Introduced an Account Association API so IVR/chat clients resolve the intended account before requesting enriched details.

3 MAKE ORCHESTRATION EXPLICIT

Replaced an opaque event-trigger chain with Step Functions, upstream checks, bounded retry, failure branching, and SNS alerts.

4 SCALE THE DATA PATH

Resolved convergent-table joins with SQL/CTEs and moved a Pandas bottleneck to PySpark across 423+ mapped fields.

MY CONTRIBUTION

ARCHITECTURE OWNERSHIP

Redesigned customer interaction, identifier cardinality, serving model, orchestration, and cache-refresh strategies.

HANDS-ON DELIVERY

Reverse-engineered source semantics, authored mappings and critical SQL/PySpark logic, and unblocked external delivery.

PRODUCTION READINESS

Defined API behavior, logging, validation, TTL reconciliation, retries, alerts, and safe reload patterns.

REPORTED PRODUCTION OUTCOMES

~500K

subscriber records per daily cycle

<200 ms

API responses observed in supported workloads

50-75 → <10 min

reported Pandas-to-PySpark runtime improvement

Observed during the period supported; production telemetry and customer data are intentionally excluded from this public portfolio.